Dominant Eigenvalue and Stable Age Distribution

In this assignment, you will analyze the Leslie matrix for female winter flounder to determine the long-term population growth factor, compute the stable age distribution, and compare the model predictions with observed age compositions.

Learning Objectives

After completing this assignment, you should be able to

- compute and interpret the eigenvalues of a Leslie matrix;
- identify the dominant eigenvalue;
- compute and normalize the corresponding right eigenvector;
- interpret the stable age distribution;
- compare a model-based age distribution with observed data;
- quantify model discrepancy using RMSE.

Background

The female population is divided into six age classes:

$$\text{age-1},\ \text{age-2},\ \text{age-3},\ \text{age-4},\ \text{age-5},\ \text{age-6+}.$$

Let

$$\mathbf{x}_k = [x_{k,1}, x_{k,2}, x_{k,3}, x_{k,4}, x_{k,5}, x_{k,6+}]^T$$

denote the age-structured population vector in year $k(k = 0, 1, 2, ..., n)$.

A one-year Leslie matrix model has the form

$$\mathbf{x}_{k+1} = L\mathbf{x}_k, \quad k = 0, 1, ..., n-1$$

where L is a nonnegative Leslie matrix. After n yearly time steps,

$$\mathbf{x}_n = L^n\mathbf{x}_0.$$

For a Leslie matrix with a simple dominant eigenvalue, repeated multiplication by L causes the population vector to become aligned with the right eigenvector associated with the dominant eigenvalue.

If $L\mathbf{w} = \lambda\mathbf{w}$, then λ is the dominant eigenvalue and represents the long-term annual population growth factor; the normalized right eigenvector $\mathbf{w}$ gives the stable age distribution.

Interpretation of λ:

$$\begin{aligned} \lambda > 1 &\implies \text{long-term growth,} \\ \lambda < 1 &\implies \text{long-term decline,} \\ \lambda = 1 &\implies \text{constant long-term size.} \end{aligned}$$

1 Enter or Import the Leslie Matrix

Use the Leslie matrix for female winter flounder.

```
import pandas as pd
import numpy as np

# Define Leslie matrix L
L = np.array([
    [0, 0.036, 0.204, 0.255, 0.144, 0.118],
    [0.789, 0, 0, 0, 0, 0],
    [0, 0.860, 0, 0, 0, 0],
    [0, 0, 0.971, 0, 0, 0],
    [0, 0, 0, 0.523, 0, 0],
    [0, 0, 0, 0, 0.750, 0.]
])
```

Check that

```
print(L.shape)
print(L)
```

returns a 6×6 matrix.

2 Compute the Dominant Eigenvalue

Use NumPy to compute all eigenvalues and right eigenvectors.

```
eigvals, eigvecs = np.linalg.eig(L)

print("Eigenvalues:")
print(eigvals)
```

Complete the following code.

```
# TODO:
# Identify the eigenvalue with the largest magnitude.

i_max =

lambda =

print("Dominant eigenvalue:", lambda)
```

Questions

1. Is the dominant eigenvalue real and positive?
2. Does the model predict long-term growth, decline, or stability?
3. Why is the dominant eigenvalue more important for long-term behavior than the remaining eigenvalues?

3 Compute the Stable Age Distribution (SAD)

Extract the right eigenvector associated with the dominant eigenvalue.

```
# TODO:
# Extract the eigenvector associated with lambda1.

stable_vector =

# Retain its real part.
stable_vector = stable_vector.real

# If necessary, reverse its sign so entries are positive.
if stable_vector.sum() < 0:
    stable_vector = -stable_vector

# TODO:
# Normalize so that the entries sum to 1.

stable_dist =
```

Verify that

$$\sum_{i=1}^{6} w_i = 1.$$

Display the results in a table.

```
age_cols = [
    "Age1", "Age2", "Age3",
    "Age4", "Age5", "Age6+"
]

sad_table = pd.DataFrame({
    "Age class": age_cols,
    "Stable proportion": stable_dist
})

print(sad_table)
```

Questions

1. Which age class has the largest stable proportion?
2. Why is the right eigenvector normalized?

3. What does “stable” mean in the phrase stable age distribution?
4. Does a stable age distribution imply that total abundance is constant?

4 Visualize the Stable Age Distribution

Create a bar chart.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5))

# TODO:
# Plot the stable age distribution as a bar chart.

plt.xlabel("Age class", fontsize=16)
plt.ylabel("Proportion", fontsize=16)
plt.xticks(fontsize=14)
plt.yticks(fontsize=14)
plt.tight_layout()

plt.savefig(
    "stable_age_distribution.pdf",
    bbox_inches="tight"
)

plt.show()
```

5 Compute Observed Age Distributions

Read the all-female counts and tow-count data.

```
excel_file = "data.xlsx"

allfem_counts = pd.read_excel(
    excel_file,
    sheet_name="AllFemales",
    index_col=0
)

tow_counts = pd.read_excel(
    excel_file,
    sheet_name="TowCount",
    index_col=0
)

allfem_counts.index = allfem_counts.index.astype(int)
tow_counts.index = tow_counts.index.astype(int)

allfem_df = allfem_counts.div(
    tow_counts["Tows"],
    axis=0
)
```

Define a function that computes the normalized average age distribution over a selected period.

```
def compute_observed_dist(df, start_year, end_year):
    """
    Compute the normalized average CPUE distribution
    over a specified interval of years.
    """

    # TODO:
    # Select the requested years and age columns.

    df_period =

    # TODO:
    # Compute mean CPUE by age class.

    avg_cpue =

    # TODO:
    # Normalize so the entries sum to 1.

    obs_dist =

    return obs_dist
```

Compute

- the observed average age distribution for 1992–2002;
- the observed average age distribution for 1992–2023.

```
obs_1992_2002 = compute_observed_dist(
    allfem_df,
    1992,
    2002
)

obs_1992_2023 = compute_observed_dist(
    allfem_df,
    1992,
    2023
)
```

6 Compare Predicted and Observed Distributions

For two vectors **a** and **b**, define the root mean squared error (RMSE)

$$\mathrm{RMSE}(\mathbf{a}, \mathbf{b}) = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(a_i - b_i)^2}.$$

Complete the function.

```
def rmse(a, b):
    # TODO:
    # Compute the root mean squared error for n=6.
    return
```

Then compute

```
rmse_1992_2002 = rmse(
    stable_dist,
    obs_1992_2002
)

rmse_1992_2023 = rmse(
    stable_dist,
    obs_1992_2023
)

print(rmse_1992_2002)
print(rmse_1992_2023)
```

Questions

1. Which observed period is closer to the model's stable age distribution?
2. Why might the observed age distribution differ from the stable age distribution?
3. How could changing survival or fecundity estimates affect the comparison?

7 Create a Comparison Figure

Create a grouped bar chart showing

- the stable age distribution;
- the observed distribution for 1992–2002;
- the observed distribution for 1992–2023.

```
x = np.arange(len(age_cols))
width = 0.25

plt.figure(figsize=(10, 6))

# TODO:
# Add the three grouped bar series.

plt.xlabel("Age class", fontsize=16)
plt.ylabel("Proportion", fontsize=16)
plt.xticks(x, age_cols, fontsize=14)
plt.yticks(fontsize=14)
plt.legend(fontsize=13)
plt.tight_layout()
```

```
plt.savefig(
    "sad_observed_comparison.pdf",
    bbox_inches="tight"
)

plt.show()
```

Interpretation

Write a short report addressing the following questions.

1. What does the dominant eigenvalue imply about long-term population growth?
2. Which age classes dominate the stable age distribution?
3. How well does the stable age distribution agree with observed age compositions?
4. What assumptions of the Leslie matrix may explain disagreement?

Deliverables

1. A completed Jupyter notebook;
2. The dominant eigenvalue of the Leslie matrix;
3. The normalized stable age distribution and its visualization;
4. A comparison of the predicted and observed age distributions, including the RMSE and a grouped bar chart;
5. A one-page interpretation of the results.